From Neural Computation to Multimodal Remote Sensing Intelligence

A Systematic, Data-Driven Review of Licheng Jiao's Research Landscape, 1998–2026

Licheng Jiao Research Atlas Project

Data snapshot: 28 July 2026

Abstract

This review maps the scholarly landscape associated with Licheng Jiao using 1,827 bibliographic records spanning 1998–2026. The corpus is anchored to the DBLP author profile 40/3714 and DOI-linked public metadata. We introduce a reproducible hierarchy of seven broad domains and 48 observed research tasks, reconstruct four phases of research evolution, identify corpus-scoped first appearances at selected journals and conferences, and analyze collaboration and Crossref Cited-by evidence. The synthesis suggests a durable research logic connecting structure-aware representation, adaptive search, intelligent perception, and domain-grounded interpretation. Contemporary multimodal and foundation-model work is therefore interpreted as a recombination of earlier methodological commitments rather than an isolated turn. Evidence boundaries remain explicit: this release is a bibliometric and title-level systematic mapping, not a substitute for source-verified full-text review.

1 Introduction

Long research careers are difficult to summarize with a short list of "representative papers." Such lists are readable, but they hide continuity, collaboration, and the movement of methods across application domains. This review instead treats the publication record as a longitudinal research system. Its central question is how a program rooted in neural computation, evolutionary search, and multiscale signal analysis expanded into visual intelligence, remote-sensing interpretation, and contemporary multimodal models.

The presentation adopts the useful discipline of modern systems surveys: define the object and scope, disclose the corpus protocol, provide an overview map before detailed taxonomy, reconnect categories through cross-cutting synthesis, and close with testable open questions [12, 19]. The visual language is original and generated from this corpus; the cited surveys serve as organizational references, not sources for copied figures or claims about Licheng Jiao's work.

1.1 Contributions

This review makes four contributions.

1. **Reproducible corpus construction.** DBLP identity anchoring, DOI linkage, public field allowlists, and versioned generation scripts make every reported count inspectable.

2. **Hierarchical systematic mapping.** Seven broad domains and 48 observed tasks connect 1,827 records while keeping the title-level coding rules visible.

3. **Longitudinal synthesis.** Four phases and selected first-venue milestones show when methodological and institutional frontiers enter the public corpus.

4. **Relational evidence.** Collaboration, venue, and Crossref citation layers complement topic counts and establish questions for a later, source-verified full-text review.

2 Corpus and Review Protocol

2.1 Identity anchor and unit of analysis

The corpus uses the DBLP author profile `40/3714` [6] as its identity anchor. DOIs and DBLP keys support record-level verification. The snapshot contains 1,827 records from 1998 through 2026, including 1,764 DOI-bearing entries. Counts refer to bibliographic records rather than deduplicated intellectual works; conference, preprint, and journal versions may therefore coexist.

Table 1: Corpus summary for the July 2026 snapshot.

Measure	Value	Measure	Value
Bibliographic records	1,827	Journal records	1,369
Conference records	453	Proceedings records	5
Records with a DOI	1,764	Observed year range	1998–2026
Distinct coauthors	2,163	Distinct venues	260
Broad domains	7	Observed fine tasks	48

2.2 Collection, inclusion, and exclusion

Records are included when they are traceable to the selected DBLP profile. Bibliographic fields are normalized into a strict public schema: stable ID, title, authors, year, publication type, venue, pages or volume, DOI, DBLP key, and public URLs. Local download state, credentials, machine paths, private annotations, and raw working logs are excluded. This separation protects the public website boundary and prevents operational metadata from being mistaken for scholarly evidence.

2.3 Hierarchical coding protocol

Each title and venue string is processed by ordered, human-readable keyword rules. The first level assigns one of seven broad domains: frontier models and multimodality, remote-sensing interpretation, visual perception, evolutionary optimization, neural and machine learning, signal and computational imaging, or interdisciplinary methods and applications. The second level assigns a more specific task, such as SAR interpretation, hyperspectral classification, change detection, multiobjective optimization, neural architecture search, segmentation, tracking, image restoration, or vision–language prompting.

This procedure favors auditability over opaque clustering. It is deterministic and easy to challenge, but a single-label taxonomy necessarily compresses hybrid work. Counts should therefore be read as navigational coordinates, not as an official classification supplied by the researcher.

2.4 Citation, collaboration, and first-venue measures

Citation counts are retrieved by DOI from Crossref Cited-by [5]. Crossref counts depend on references deposited into its network and may under-report the complete citation record. Coauthor counts represent record-level co-occurrence after name normalization; they do not measure contribution, seniority, or supervision.

Selected venue milestones are generated by exact or documented venue-family matching. "First" means the earliest matching year in this DBLP-anchored corpus. When multiple records share that first year, all are grouped rather than assigned a false within-conference chronology.

Evidence boundary. The current public catalog does not contain source-verified abstracts or full text for the complete corpus. This review supports bibliometric and title-level systematic mapping; it does not infer experimental superiority, numerical performance, or technical novelty from titles alone.

3 A Unifying View of the Research Program

The corpus supports a four-layer interpretation of the research program. *Structure-aware representation* describes repeated attention to multiscale, sparse, spectral–spatial, geometric, and attentive representations. *Adaptive search and optimization* captures evolutionary, multiobjective, transfer, and architecture-level search. *Intelligent perception* includes classification, detection, segmentation, tracking, restoration, and matching. Finally, *domain-grounded interpretation* reflects sustained engagement with SAR, PolSAR, hyperspectral, aerial, and satellite data.

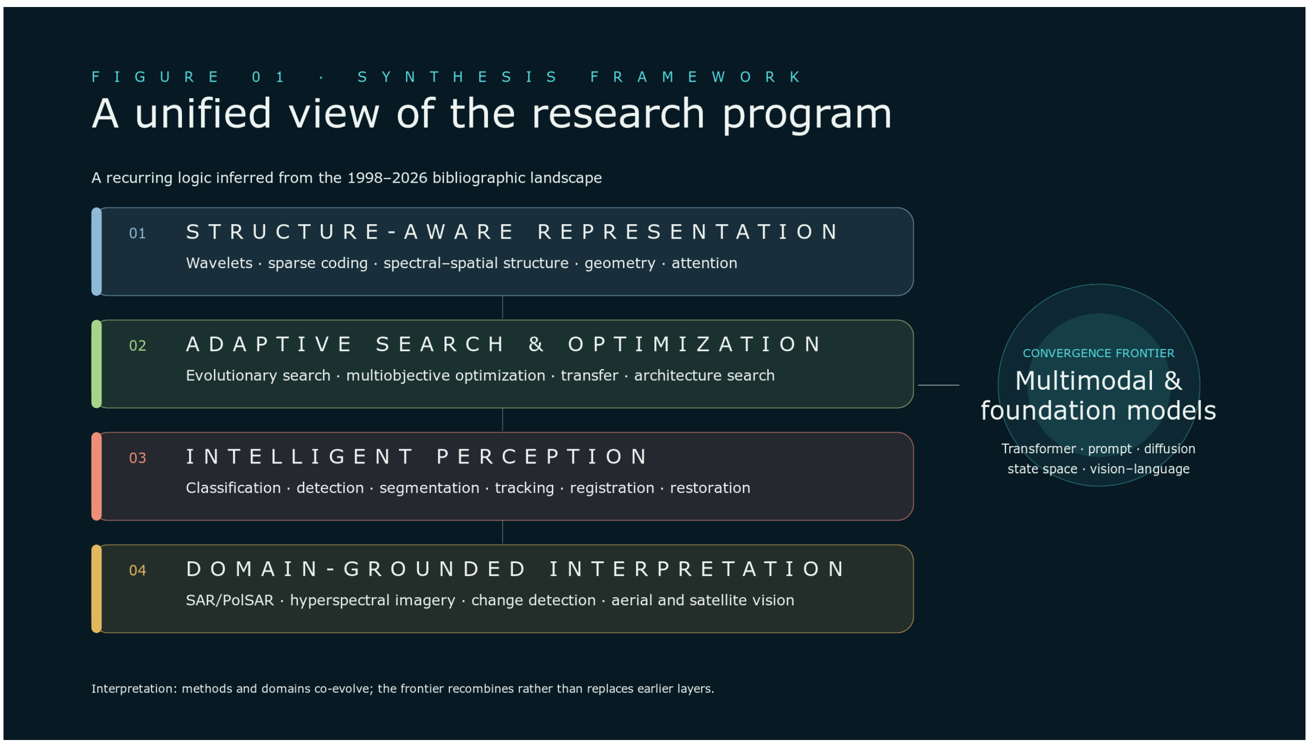

Figure 1: Original synthesis framework for the research program. The diagram summarizes recurring methodological relations inferred from the bibliographic landscape; it is not an author-supplied taxonomy.

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These layers are not historical stages. They interact across time: remote-sensing constraints motivate new representations; search procedures choose features and architectures; perception tasks provide demanding evaluation settings; and modern multimodal systems recombine all four. The framework therefore interprets the contemporary frontier as convergence rather than replacement.

3.1 Structure-aware representation

Early multiscale and wavelet work, sparse representations, spectral–spatial modeling, geometric matching, attention, and recent state-space architectures share a concern with how structure is represented. The techniques differ substantially, but the bibliographic trajectory repeatedly returns to the premise that signals and images should be modeled through meaningful relations—across scale, spectrum, space, geometry, or time.

3.2 Adaptive search and optimization

Evolutionary computation is not confined to a historical period. It appears in multiobjective search, feature and band selection, scheduling, clustering, architecture search, and large-scale optimization. Highly cited examples such as MOEA/D with adaptive weight adjustment [17], Two_Arch2 [21], and decision-variable analysis for large-scale optimization [15] illustrate the visibility of this methodological axis.

3.3 Intelligent perception and domain grounding

Classification, detection, segmentation, tracking, matching, registration, and restoration form the perception layer. Remote sensing supplies a sustained domain context whose multiscale, multimodal, and physically grounded data challenge generic vision assumptions. Change detection in SAR imagery [9], spectral–spatial hyperspectral modeling [25], and image registration [14] are prominent examples in the citation landscape.

4 Longitudinal Evolution and Venue Frontiers

4.1 Four phases

The publication trajectory can be summarized in four phases.

1998–2007: 179 records Neural networks, evolutionary computation, clustering, wavelets, and early pattern recognition establish a methodological base.

2008–2014: 293 records Multiobjective optimization, sparse representation, imaging, SAR, and remote-sensing classification become increasingly connected.

2015–2020: 530 records Deep visual learning expands across hyperspectral data, change detection, scene recognition, detection, segmentation, and multimodal fusion.

2021–2026: 825 records Transformers, prompts, vision–language interaction, diffusion, Mamba, state-space models, and foundation-model adaptation recombine earlier themes.

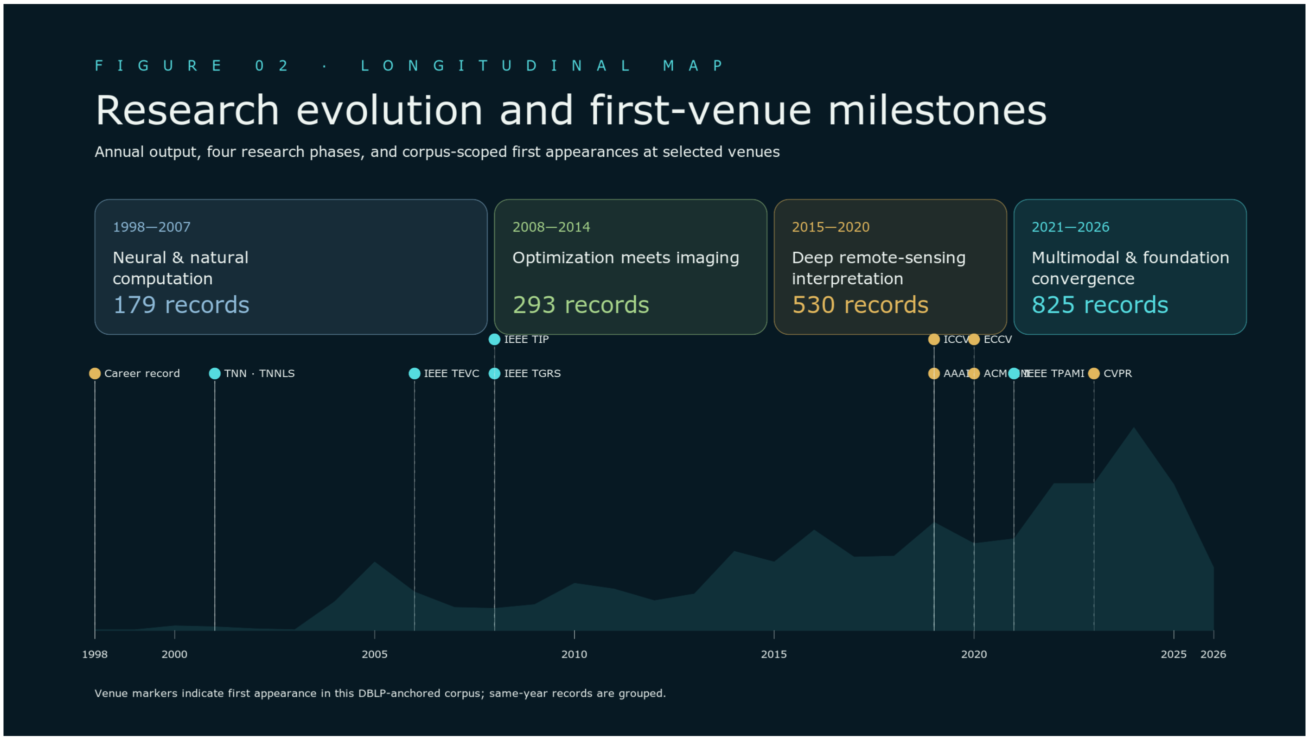

Figure 2: Research evolution and selected first-venue milestones. Markers denote the first appearance in the current DBLP-anchored corpus, not a claim about records outside the dataset.

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4.2 Selected corpus-scoped first appearances

Table 2 records selected venue coordinates. The focus is not prestige ranking. The ledger makes institutional expansion visible and gives readers concrete entry points into the bibliography.

Table 2: Selected first appearances in the current DBLP-anchored corpus.

Year	Venue coordinate	First-year record or records
1998	Corpus start	Total least mean squares algorithm
2001	IEEE TNN / TNNLS	Multiwavelet neural network and its approximation properties
2002	Pattern Recognition	Linear programming support vector machines
2005	NeurIPS / NIPS	Response Analysis of Neuronal Population with Synaptic Depression

Year	Venue coordinate	First-year record or records
2006	IEEE TEVC	An organizational coevolutionary algorithm for classification
2008	IEEE TIP	Wavelet-Based Bayesian Image Estimation
2008	IEEE TGRS	Spectral Clustering Ensemble Applied to SAR Image Segmentation
2019	AAAI	Multi-Precision Quantized Neural Networks via Encoding Decomposition of {-1,+1} [20]
2019	ICCV	AFD-Net; and Better and Faster: Exponential Loss for Image Patch Matching [18]
2019	IJCAI	Accelerated Incremental Gradient Descent using Momentum Acceleration with Scaling Factor
2020	ACM MM	A Novel Object Re-Track Framework for 3D Point Clouds [7]
2020	ECCV	Supervised Edge Attention Network for Accurate Image Instance Segmentation [4]
2021	IEEE TPAMI	Accelerated Variance Reduction Stochastic ADMM for Large-Scale Machine Learning [13]
2023	CVPR	Curvature-Balanced Feature Manifold Learning; and an online self-training study for semantic segmentation [16, 24]
2023	ICLR	Delving into Semantic Scale Imbalance

5 Hierarchical Thematic Architecture

Figure 3: Hierarchical thematic map. Task colors are variations within each domain family, matching the interactive paper graph. Counts are title-level, single-label coordinates.

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5.1 Remote-sensing interpretation

Remote sensing is the largest single-label domain with 667 records. Its internal tasks include scene and land-cover classification, hyperspectral and multispectral learning, SAR/PolSAR, change detection, object detection and tracking, registration and fusion, and remote-sensing captioning. The count is large enough to demonstrate a persistent domain core, but the cross-layer framework cautions against treating it as an isolated application silo.

5.2 Interdisciplinary methods

The interdisciplinary domain contains 434 records in communications, medical and bioinformatics applications, data mining, mathematical methods, security, 3D geometry, multimedia, and other systems. This category is intentionally conservative: it receives records for which another broad domain is not the strongest title-level coordinate. It therefore combines genuine interdisciplinary reach with residual classification uncertainty.

5.3 Evolutionary optimization

Evolutionary and intelligent optimization contributes 236 records. The task hierarchy separates multiobjective and many-objective optimization, general evolutionary algorithms, swarm intelligence, neural architecture search, clustering, transfer and multitask evolution, and combinatorial optimization. The domain's influence is broader than the standalone count because search methods reappear inside feature selection, architecture design, and remote sensing.

5.4 Visual perception

Visual perception contains 198 records. Segmentation is the largest assigned task, followed by recognition and classification, tracking and re-identification, detection, video understanding, matching and localization, and medical vision. Related visual work is also assigned to remote sensing and frontier models, so this count is a lower bound on visual perception's role in the full corpus.

5.5 Frontier, learning, and imaging methods

Frontier-model work contains 149 records, including Transformer and attention architectures, multimodal learning, diffusion and generative models, vision–language prompting, state-space models, and foundation models. Neural and machine learning contributes another 73 records focused on self-, semi-, and unsupervised learning; few-shot and incremental learning; representation learning; and domain adaptation. Signal and computational imaging contributes 70 records in wavelets, sparse representations, reconstruction, super-resolution, restoration, denoising, and fusion.

Table 3: Broad thematic domains produced by the transparent title-level taxonomy.

Domain	Records
Remote-sensing interpretation	667
Interdisciplinary methods and applications	434
Evolutionary and intelligent optimization	236
Visual perception and understanding	198
Frontier models and multimodality	149
Neural and machine learning	73
Signal and computational imaging	70

6 Collaboration and Venue Ecology

The corpus contains 2,163 normalized coauthor names. The highest collaboration frequencies are Fang Liu (433 records), Shuyuan Yang (297), Lingling Li (249), Xiangrong Zhang (245), Xu Liu (238), Biao Hou (237), Ronghua Shang (177), Shuang Wang (158), Wenping Ma (138), and Yangyang Li (135).

These repeated relationships connect optimization, remote sensing, imaging, and visual learning, but record counts cannot determine contribution or supervision.

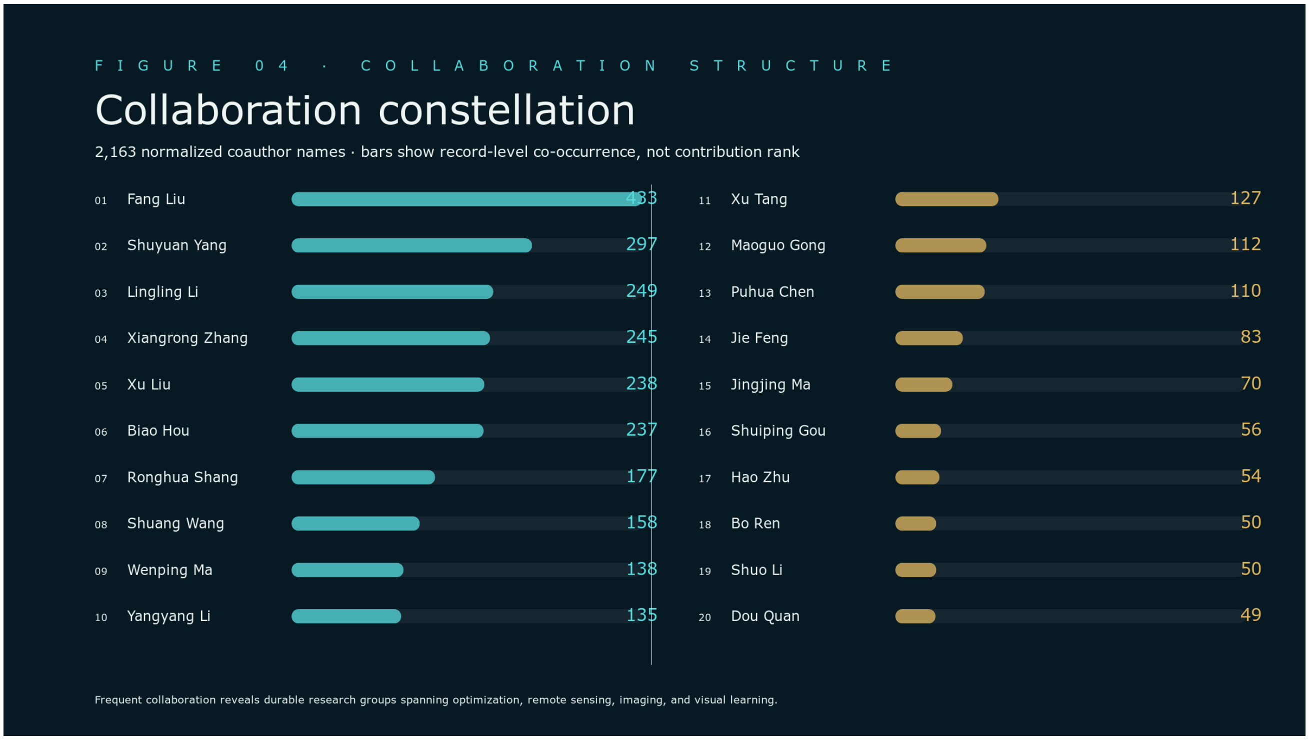

Figure 4: Top twenty coauthor names by record-level co-occurrence. The complete, searchable list is available in the interactive Atlas.

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Venue concentration reinforces the thematic story. The most frequent sources are *IEEE Transactions on Geoscience and Remote Sensing* (255 records), CoRR (125), IGARSS (104), *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* (83), *IEEE Geoscience and Remote Sensing Letters* (71), *Remote Sensing* (69), *Neurocomputing* (66), and *Pattern Recognition* (63). The mixture of archival journals, major conferences, and preprints reflects both a stable domain core and rapid uptake of new methods.

7 Impact Structure

Of 1,764 DOI-bearing records, 1,682 matched the current Crossref Cited-by snapshot. Raw citation rankings favor older publications and are sensitive to database coverage. The Atlas preserves the raw count while adding a publication-year cohort layer. A record is highlighted when it reaches the top decile of its year cohort with at least five citations. A stronger landmark layer requires the year-specific 99th percentile and at least 100 citations.

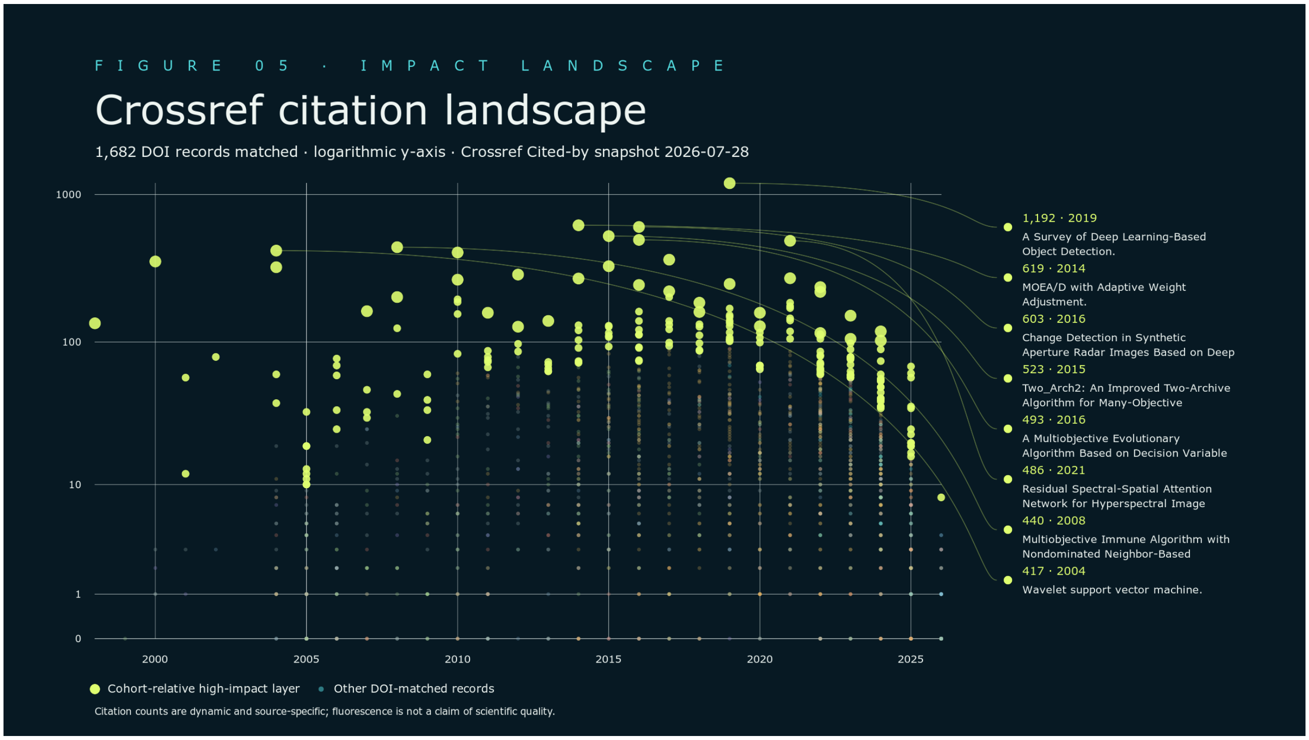

Figure 5: Crossref citation landscape. Fluorescence identifies unusually visible records relative to their publication-year cohort; it is not a claim of scientific quality.

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Table 4: Highest Crossref Cited-by counts among DOI-matched corpus records, updated 28 July 2026. Counts are dynamic and source-specific.

Citations	Year	Record
1,192	2019	A Survey of Deep Learning-Based Object Detection [11]
619	2014	MOEA/D with Adaptive Weight Adjustment [17]
603	2016	Change Detection in Synthetic Aperture Radar Images Based on Deep Neural Networks [9]
523	2015	Two_Arch2: An Improved Two-Archive Algorithm for Many-Objective Optimization [21]
493	2016	A Multiobjective Evolutionary Algorithm Based on Decision Variable Analyses for Problems With Large-Scale Variables [15]

Citations	Year	Record
486	2021	Residual Spectral-Spatial Attention Network for Hyperspectral Image Classification [25]
440	2008	Multiobjective Immune Algorithm with Nondominated Neighbor-Based Selection [8]
417	2004	Wavelet Support Vector Machine [23]
405	2010	Adaptive NN Backstepping Output-Feedback Control for Stochastic Nonlinear Strict-Feedback Systems With Time-Varying Delays [3]
362	2017	Remote Sensing Image Registration With Modified SIFT and Enhanced Feature Matching [14]
352	2000	A Novel Genetic Algorithm Based on Immunity [10]
327	2015	A Novel Point-Matching Algorithm Based on Fast Sample Consensus for Image Registration [22]

Citation counts should never be treated as a direct measure of scientific quality. Their strongest role here is navigational: they identify records that warrant close reading and help compare visibility across topics and years.

8 Cross-Cutting Synthesis

Three recurring mechanisms reconnect the thematic map.

1. **Structure-aware representation.** Multiscale transforms, sparse representations, spectral–spatial relations, geometric constraints, attention, and state-space models all encode structure rather than treating feature learning as undifferentiated.

2. **Search, selection, and adaptation.** Evolutionary search, multiobjective optimization, feature selection, transfer, prompting, and architecture search repeatedly address how a model or solution process should adapt to a task.

3. **Method–domain coevolution.** Remote sensing is not merely an application destination. Its multiscale, multimodal, geometric, and physically grounded problems repeatedly motivate new representations and optimization formulations.

These mechanisms explain why the contemporary frontier is better described as accumulation and recombination than as disconnected trend following. Transformer, vision–language, diffusion, prompt, Mamba, and foundation-model papers appear inside a program already organized around structured signals, adaptive search, and demanding perception domains.

9 Open Questions

The current map motivates five questions for a deeper full-text review.

1. **How does domain structure enter modern foundation models?** Source-level analysis should distinguish architectural encoding, data construction, objectives, prompting, and post-training adaptation.

2. **Where does evolutionary search create unique value?** Comparative reading is needed to separate shared mechanisms from inherited terminology across classical evolutionary computation, architecture search, and prompt or model adaptation.

3. **How do representations move across sensors and tasks?** Version-linked experiments could reveal when representations developed for SAR, hyperspectral, change detection, or matching transfer to other modalities.

4. **Can impact be measured beyond raw citations?** Future releases should combine source-verified references, citation contexts, benchmark adoption, code availability, and field-normalized indicators.

5. **How does collaboration mediate research evolution?** Affiliation-aware and author-disambiguated analysis could test how methods, domains, and venue frontiers travel across teams and generations.

10 Limitations and Reproducibility

The evidence base supports corpus mapping, descriptive bibliometrics, title-level thematic coding, collaboration counts, corpus-scoped venue firsts, and source-labeled citation analysis. It does not support claims about experimental results, causal influence, intellectual priority, or a paper's technical contribution beyond verified metadata.

Conference, preprint, and journal versions may coexist. Author-name normalization cannot resolve every homonym. Crossref coverage is incomplete and dynamic. A deterministic one-label taxonomy can misclassify hybrid work. "First" means first appearance in the selected DBLP-anchored dataset, not an absolute career claim.

Reproducibility assets include the public field allowlist, taxonomy rules, milestone generator, citation-refresh script, figure generator, QR assets, semantic HTML, this LaTeX manuscript, and the compiled PDF. The analysis can therefore be extended incrementally as verified abstracts, full text, references, and affiliations become available. The HTML and LaTeX follow arXiv's guidance on semantic structure and alternative descriptions [1, 2].

11 Conclusion

The 1998–2026 record portrays a research program with unusual continuity across neural computation, evolutionary optimization, image intelligence, and remote sensing. Its contemporary work on multi-modal and foundation models is not an isolated turn; it extends longstanding interests in structured representation, adaptive search, and complex visual signals. The open Atlas and this systematic map provide a transparent baseline for source-verified technical synthesis, reference-network analysis, and reusable scholarly tools.

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